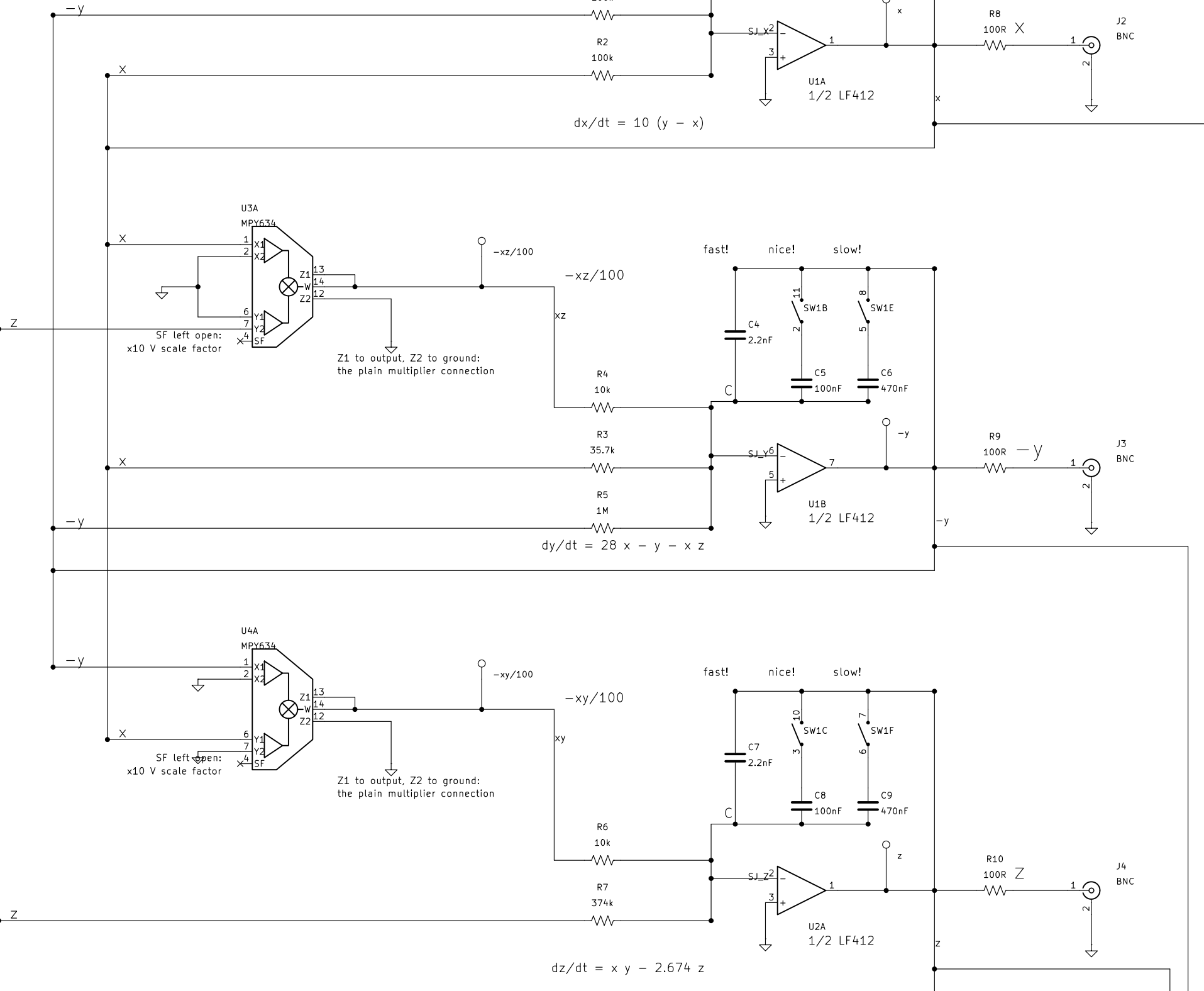


C = 0.47 uF (slow!)
= 0.1 uF (nice!)
= 2000 pF (fast!)

...built here as 2.2 nF always fitted, plus 100 nF and 470 nF switched in by SW1. See the notes at the right of the sheet.



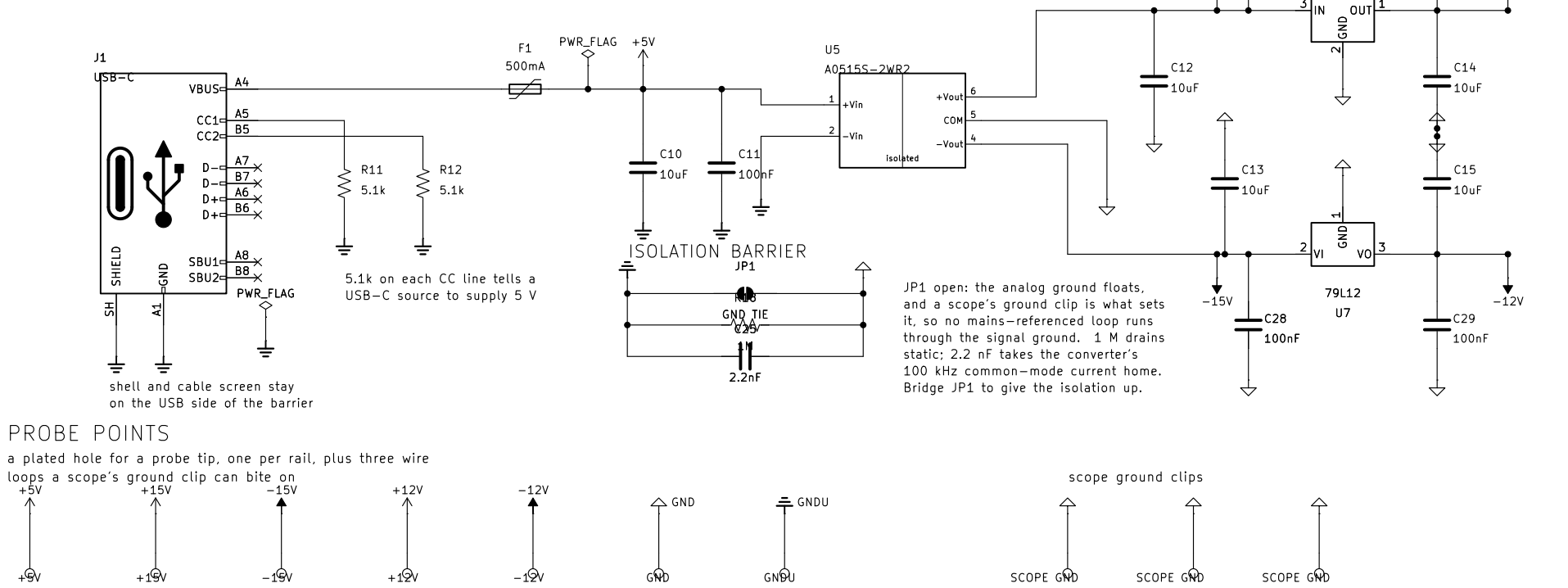
CHAOS LAMP

One lamp, three colours, driven from the three outputs.

$-12\text{ V} \times 4.7\text{ k} / (4.7\text{ k} + 33\text{ k}) = -1.5\text{ V}$, buffered by U2B -- the half of the LF412 Paul never needed -- and held on the lamp's common cathode. Putting the cathode below ground is what lets a signal that swings either side of zero switch a colour on and off; each one lights only above its own turn-on voltage. The taps are on the output side of each integrator, ahead of the 100 ohm series resistor, so no lamp current flows in the BNC output impedance.

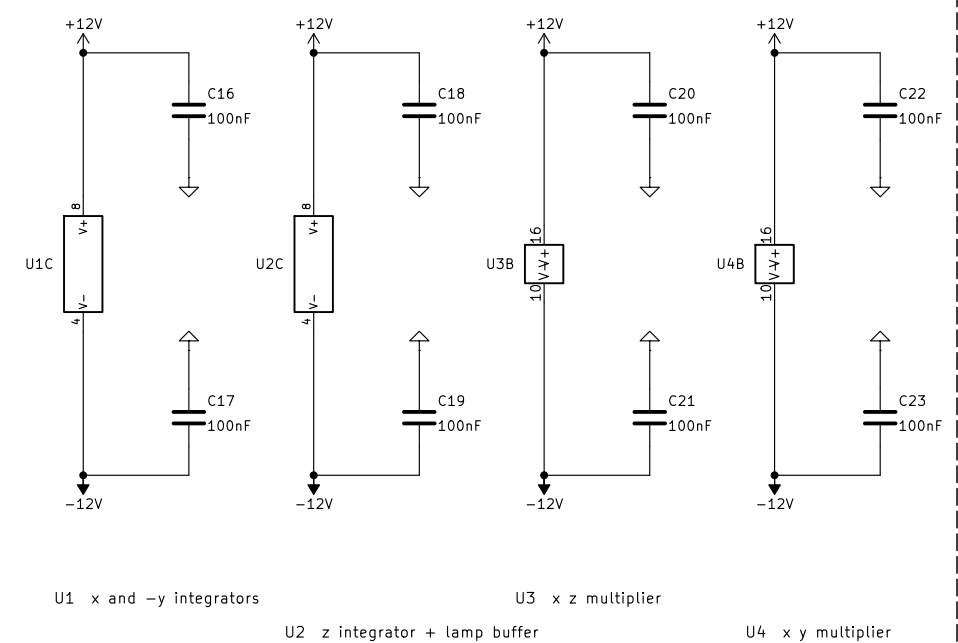
POWER: USB-C 5 V in, +/-12 V out

about 25 mA per rail; the 2 W module is rated +/-66 mA

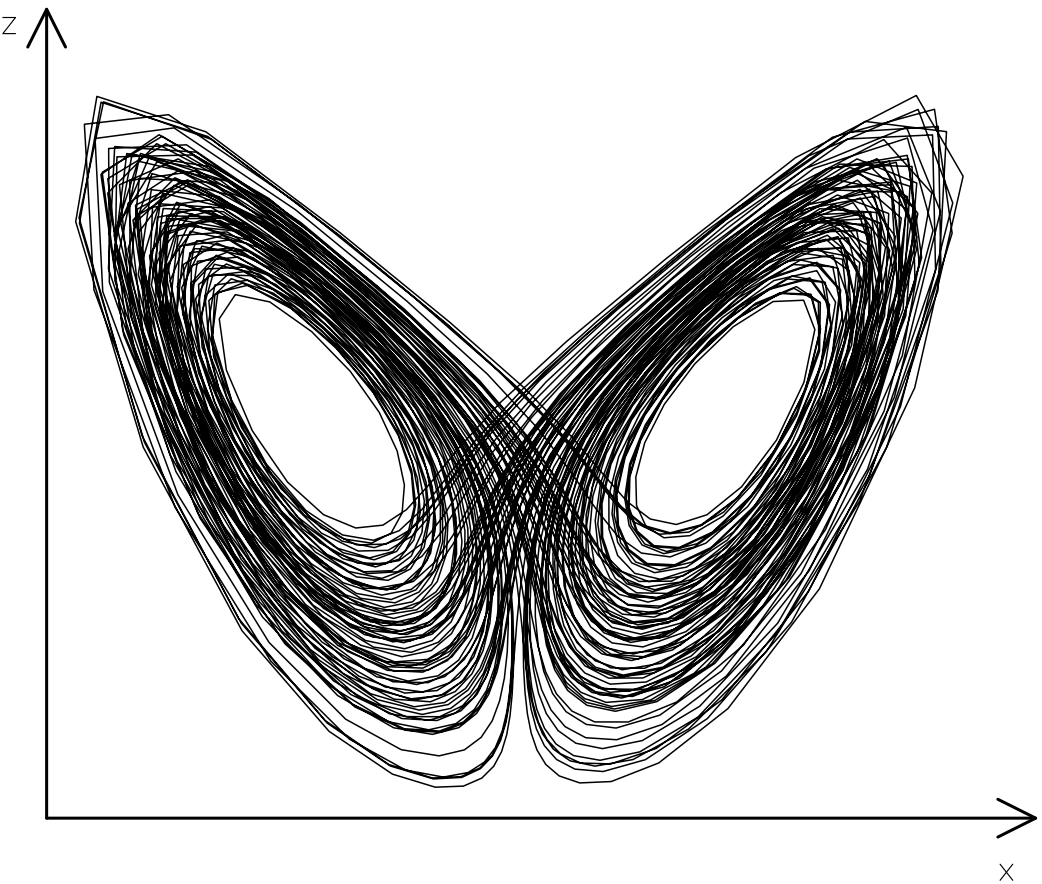


SUPPLY PINS AND BYPASSING

one 0.1 uF at every supply pin, right at the package



Hook x and z to a scope in X-Y and this is what you get: the attractor's characteristic owl's face. The trace below is the real solution of these equations, integrated at s = 10, r = 28, b = 8/3.



HOW THIS CIRCUIT SOLVES THE EQUATIONS

$$\begin{aligned} dx/dt &= s(y - x) & s &= 10 \\ dy/dt &= r x - y - x z & r &= 28 \\ dz/dt &= x y - b z & b &= 8/3 \end{aligned}$$

Each op-amp is an integrator with the terms of one derivative summed at its inverting input, so the weight of every term is simply 1 MEG / R :

$$\begin{aligned} R2 = R1 = 100\text{ k} &\rightarrow 1\text{ M}/100\text{ k} = 10.00 = s && \text{(and the } -x \text{ damping)} \\ R3 = 35.7\text{ k} &\rightarrow 1\text{ M}/35.7\text{ k} = 28.01 = r \\ R5 = 1\text{ M} &\rightarrow 1\text{ M}/1\text{ M} = 1.00 = \text{the } -y \text{ term} \\ R7 = 374\text{ k} &\rightarrow 1\text{ M}/374\text{ k} = 2.674 = b && (8/3 = 2.667, 0.3 \% \text{ high}) \\ R4 = R6 = 10\text{ k} &\rightarrow 1\text{ M}/10\text{ k} = 100 && \text{cancels the multipliers' } /100 \end{aligned}$$

Voltages carry the dimensionless variables at 0.1 V per unit, so x swings +/-2 V, y +/-2.7 V and z runs 0 to 4.8 V. An MPY634 forms A*B/10, so the product of two 0.1 V/unit signals comes out 100x too small -- which is exactly what the 10k weighting resistors R4 and R6 put back. The minus signs on both products are free: U3 and U4 simply have their differential inputs swapped.

The time scale is tau = 1 MEG x C, and C is switchable:

speed	1	2	3	4	5	6	C	tau = 1M x C
fast!	off	off	off	off	off	off	2.2 nF	2.2 ms
nice!	ON	ON	ON	off	off	off	102 nF	102 ms
slow!	off	off	off	ON	ON	ON	472 nF	472 ms

Never both banks ON: that is 572 nF, which works but is not one of the three settings the board is labelled for.

Paul's original ran on a bench +/-15 V supply and used 2000 pF, 0.1 uF and 0.47 uF. Here the three values are built up in parallel because no supplier stocks a 3-pole 3-position switch; 2.2 nF is always fitted.

MOUNTING

Four M3 holes, one per board corner. No electrical function, but with three BNCs along one edge the cables lever on the board, so it wants standoffs rather than a bare bench.



Open source -- https://github.com/gallicchio/al_unleashes_chaos

Every coefficient is 1 MEG / R: signals are 0.1 V per unit

s = 10, r = 28, b = 8/3

dx/dt = s(y-x) dy/dt = rx - y - xz dz/dt = xy - bz

Circuit by Paul Horowitz | PCB by Jason Gallicchio and Claude

Sheet: /

File: lorenz.kicad_sch

Title: **Al Unleashes Chaos -- Lorenz Attractor**

Size: A2 Date: 2026-09-15

Rev: A

KiCad E.D.A. 10.0.6

Id: 1/1